A Holomorphic Hénon Cycle and Transfer-Ownership Pilot

Anonymous Route-A report

Abstract

We freeze the complex polynomial automorphism $F(z,w) = (w, w^2 - z/4)$ and solve its period-one and period-two equations exactly. Lefschetz-type Jacobian weights give the trace values $L_1 = 0$ and $L_2 = -1664/1725$, hence a formal determinant prefix $1 + 832z^2/1725$. The inverse pullback has coordinate degrees $2, 4, 8$ after three iterates, which is a precise warning that finite polynomial Galerkin truncations do not by themselves supply a source-native Fredholm owner. This is an exact finite A2 prefix; complete complex coding, nuclearity, and a positive prime-like clock remain open.

1 Frozen map and cycles

Let

$$F(z,w) = (w, w^2 - \tfrac{1}{4}z), \qquad DF(z,w) = \begin{pmatrix} 0 & 1 \\ -1/4 & 2w \end{pmatrix}.$$

The fixed-point equation is $x(x - 5/4) = 0$. Eliminating y from $y = x^2 - x/4$ and $x = y^2 - y/4$ gives

$$\mathrm{Res}_y = -x(4x-5)(4x^2+3x+3)/16.$$

Thus F^2 has four fixed points, with one complex-conjugate primitive two-cycle. For a fixed point of F^n, use the exact weight $1/\det(I - DF^n)$.

2 Exact trace prefix

The two fixed points of F have denominators $5/4$ and $-5/4$, so their weights cancel and $L_1 = 0$. At the four points of F^2 the denominators are $25/16$, $-75/16$, and $-23/16$ twice. Consequently

$$L_2 = \frac{16}{25} - \frac{16}{75} - \frac{32}{23} = -\frac{1664}{1725}.$$

The formal trace-to-determinant relation gives

$$D(z) = \exp\left(-L_1 z - \frac{L_2}{2} z^2 + O(z^3)\right) = 1 + \frac{832}{1725} z^2 + O(z^3).$$

Proposition 1. *The resultant, the four period-two points, the two weighted traces, and the displayed determinant coefficient are independently reproduced by the exact checker and a SymPy resultant calculation.*

Proof. The checker recomputes the two elimination equations and the Jacobian products without importing producer data. The symbolic cross-check verifies the same resultant and rational coefficient. $\square$

3 Ownership obstruction

The inverse is

$$F^{-1}(z,w) = (4z^2 - 4w, z).$$

Pulling back the coordinate z raises total degree from 1 to 2, then 4, then 8. Hence the obvious finite polynomial-degree spaces are not invariant. A genuine holomorphic Fredholm construction would need a named function space, compactness/nuclearity, and a tail estimate; a finite matrix or a post-hoc diagonal cannot replace those arguments.

4 Route-A verdict and scope

The exact finite ledger earns ‘A2_CERTIFIED_PREFIX‘; complete complex primitive-orbit coding and a positive intrinsic clock are not proved, so ‘A1_OPEN‘. We do not claim a global complex repeller, an analytic Fredholm determinant, prime correspondence, Riemann zeros, automorphy, or a Hilbert–Pólya operator. All source, checker, replay, mutation, and manifest files are distributed with the paper.